

Research paper

Machine learning-driven risk prediction and feature identification for major depressive disorder and its progression: an exploratory study based on five years of longitudinal data from the US national health survey

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ABSTRACT

Background: Major depressive disorder (MDD) presents significant public health challenges due to its increasing prevalence and complex risk factors. This study systematically analyzed data from 2019 to 2023 to explore trends in MDD incidence, symptom progression, and multidimensional influencing factors.

Objective: The study aimed to identify key risk factors and mechanisms of MDD progression through multidimensional variable analysis and machine learning model evaluation, optimizing predictive models to provide a scientific basis for early screening, targeted interventions, and public health policy formulation.

Methods: Data from the U.S. National Health Interview Survey (NHIS) were analyzed using LASSO regression for feature selection and various machine learning models (e.g., AdaBoost, Gradient Boosting, Gaussian NB, Extra Trees) for risk prediction. SHAP value analysis was employed to interpret the models and identify key predictors across psychological, social, and biological domains.

Results: Depression risk was closely associated with psychological health (e.g., PHQ82, ANXLEVEL), social support (e.g., DISCRIM1, PCNTADLT), and biological health (e.g., COGMEMDFF, DEPEV). AdaBoost, Gradient Boosting, Gaussian NB, and Extra Trees demonstrated superior performance in different tasks, with SHAP analysis highlighting the critical role of psychological health in risk prediction and the importance of social support and cognitive deficits in symptom progression.

Conclusions: This study provides a scientific foundation for early screening of MDD, personalized intervention strategies, and public health policy optimization. Future research should expand data coverage, optimize models, and incorporate causal inference methods to enhance the precision and applicability of depression risk prediction.

1. Introduction

Depression is a common mental health issue characterized by persistent feelings of sadness and a loss of pleasure, accompanied by cognitive and physical symptoms that significantly impact daily functioning and quality of life (Marx et al., 2023). In recent years, the prevalence of depression has increased markedly, particularly among younger populations. Three nationally representative surveys in the United States revealed a steady decline in adult mental health from 1993 to 2020 (Udupa et al., 2023). For individuals aged 18–25, the number of mentally unhealthy days per month rose from 3 to 6.02, and the proportion of young adults reporting poor mental health for half or more of the time doubled between 1993 and 1999 and 2018–2020. Similarly,

data from the U.S. National Health and Nutrition Examination Survey (NHANES) indicated that the number of young individuals meeting the criteria for moderate to severe depression more than doubled between 2006 and 2007 and 2017–2020, with a 50 % increase in those aged 50 and older. These trends demonstrate that depression now affects a broader demographic, extending from adolescents to young and middle-aged adults, potentially posing significant challenges to social and healthcare resources.

Major Depressive Disorder (MDD) is strongly associated with premature mortality, with high rates of comorbidity and suicide risk (McIntyre and Gill, 2020). The risk of suicide is nearly 20 times higher among individuals with MDD compared to those without, alongside elevated risks of suicidal ideation, planning, and attempts (Marx et al.,

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2023). The presence of psychotic features such as delusions or hallucinations further heightens the risk of suicide (Orsolini et al., 2020). Additional key risk factors include male gender, a history of suicide attempts, severe depressive symptoms, and comorbid anxiety or substance use disorders (Li et al., 2022b). Despite the increasing number of individuals affected by MDD, there is currently no universally accepted “gold standard” for diagnosing depression (Miller et al., 2021). Diagnosis primarily relies on mental health professionals’ evaluations combined with standardized tools such as the DSM-5 and the Patient Health Questionnaire-9 (PHQ-9) (Martin-Key et al., 2022). While the DSM-5 provides the main diagnostic criteria for depression (Forbes et al., 2024), the PHQ-9 is widely used as a screening tool to quickly assess symptom severity (Sun et al., 2020). However, these tools depend on self-reported data and professional judgment, which may lead to misdiagnosis or underdiagnosis, particularly in resource-limited settings. This underscores the urgent need for more precise and comprehensive diagnostic methods.

MDD is a complex mental disorder influenced by various biological, environmental, genetic, and neurological factors. Biologically, MDD is traditionally associated with the monoamine hypothesis, which suggests that deficits in neurotransmitters such as serotonin, norepinephrine, and dopamine in the central nervous system may contribute to the disorder (Cui et al., 2024). However, the delayed onset of antidepressant efficacy and the heterogeneity of treatment responses suggest the involvement of additional mechanisms, such as alterations in brain structure and function (Liao et al., 2024), inflammatory responses (Bhatt et al., 2023), gut-brain axis dysregulation (Riggott et al., 2024), and hypothalamic-pituitary-adrenal (HPA) axis dysfunction (Freimer et al., 2022). Genetic studies indicate that MDD has a moderate heritability ranges from 30 % to 50 %. (Kendall et al., 2021). Children of parents with MDD have twice the risk of developing the disorder compared to those without such a family history (Agerbo et al., 2021). Gene-environment interactions are significant, with environmental factors like childhood trauma influencing the risk of MDD through epigenetic mechanisms (Normann and Buttenschön, 2020). Environmental risk factors include childhood emotional or physical abuse, low socioeconomic status, and inadequate social support (Berk et al., 2023). Conversely, protective factors such as a positive family environment, extroverted personality traits, and higher self-esteem can enhance psychological resilience (Metts et al., 2021). Neurological studies have identified dysfunctions in multiple brain networks associated with MDD, particularly involving the prefrontal cortex, anterior cingulate cortex, and hippocampus (Yang et al., 2021). Structural and functional abnormalities in these areas are closely linked to the emotional, cognitive, and behavioral manifestations of MDD (Li et al., 2022a). Thus, the onset of MDD arises from a multidimensional interaction of factors, presenting significant challenges for treatment and intervention.

The application of machine learning (ML) in MDD research has grown substantially, offering significant advancements in the understanding and management of the disorder (Chen et al., 2022). In diagnosis and classification, ML can integrate clinical data, behavioral patterns, and biomarkers to accurately identify individuals with MDD and stratify symptoms, improving diagnostic efficiency and accuracy (Winter et al., 2024). In risk prediction, ML algorithms can incorporate demographic, lifestyle, and genetic data to quantify individual risks for MDD, supporting early intervention and preventive measures (Taliaz et al., 2021). For treatment evaluation, ML can precisely quantify patients’ responses to different therapeutic approaches, aiding in the optimization of personalized treatment plans (Adam et al., 2020). Furthermore, by analyzing data from social media (Hinduja et al., 2022), speech, and facial expressions, ML techniques can enable real-time monitoring of emotional changes (Gupta and Batra, 2023), providing effective support for ongoing mental health management. These applications not only advance MDD research and practice but also offer innovative pathways for precision medicine and personalized health management. This study employs a retrospective cohort design using

data from the U.S. National Health Interview Survey (NHIS) from 2019 to 2023. Machine learning algorithms are utilized to develop risk prediction models for MDD and its progression, with interpretable models (e.g., SHAP analysis) identifying core features associated with MDD. By integrating multidimensional data over five years, this research aims to explore the critical mechanisms influencing MDD, construct precise predictive models, and provide theoretical and data-driven support for personalized management and targeted interventions.

2. Methods

The U.S. National Health Interview Survey (NHIS), led by the National Center for Health Statistics (NCHS), is a nationwide health survey aimed at supporting policy-making and health research by collecting data on residents’ health status, healthcare utilization, and associated factors (Thomas, 2021). NHIS surveys approximately 30,000 adults and 9000 children annually, ensuring national representativeness. The survey includes an annual core, rotating core, sponsored content, and emerging topics, covering demographics, health insurance, medical services, mental health, physical activity, and more. Through a redesigned structure, NHIS enhances the flexibility and applicability of its data, providing valuable resources for health-related analysis and decision-making (Zablotsky et al., 2023).

2.1. Study design

This study utilized retrospective data from the U.S. National Health Interview Survey (NHIS) between 2019 and 2023, specifically focusing on the rotating core dataset. Using the variable HHX (a unique household identifier) for data merging, the study population was divided into four groups to explore risk prediction and progression factors for major depressive disorder (MDD): Group 1 analyzed 2019 features to predict 2022 depression risk, Group 2 analyzed 2020 features to predict 2023 depression risk, Group 3 marked individuals whose depression progressed from mild or moderate in 2019 to severe by 2022, and Group 4 marked individuals whose depression progressed from mild or moderate in 2020 to severe by 2023. This grouping design, grounded in a longitudinal logic of time spans and depression state progression, allows for simultaneous exploration of predictive features for depression risk and trends in condition worsening. To ensure comparability across groups and the applicability of external validation, shared variables were retained for analysis. Outliers were identified using the interquartile range (IQR) method and marked as missing values (Ibrahim et al., 2021), which were addressed using multiple imputation ($K = 5$) to minimize data bias (Chang et al., 2020). During the data standardization phase (Castiglioni et al., 2021), all variables were processed with scikit-learn’s Standard Scaler to achieve consistent feature scaling, enhancing the stability of subsequent model training. Thirteen machine learning algorithms, including logistic regression, random forest, support vector machines, and neural networks, were applied, and model performance was assessed using metrics such as AUC, precision, accuracy, recall, and F1 score (Naidu et al., 2023). The best models for each group were identified using standardized composite scores, with K-fold ($K = 5$) cross-validation to prevent overfitting (Seraj et al., 2023), and external validation sets from later years were used to evaluate generalization (Ho et al., 2020). For interpretability, SHAP (Shapley Additive Explanations) analysis was applied to each group’s best model (Nohara et al., 2022), extracting the top 10 important features and visualizing their contributions. These features included mental health indicators such as PHQ82 and ANXLEVEL, as well as cognitive and social variables like COGMEMDF and DISCRIM1, revealing key factors and potential mechanisms underlying MDD progression. Through multidimensional machine learning methods combined with external validation and SHAP analysis, this study optimized MDD risk prediction models and systematically identified critical factors influencing its progression, providing data-driven insights and theoretical support for targeted interventions.

and personalized management while offering a reference for future predictive model development and clinical applications.

2.2. Study population

The study population consisted of adult participants (age ≥ 18) from the NHIS between 2019 and 2023. In NHIS sampling, one sample adult is randomly selected from each household member registry interview. Typically, the selected adult provides self-reported information unless physical or mental limitations require a proxy respondent. The study focused on analyzing demographic characteristics, health status, mental health, and healthcare utilization data, exploring key factors and risk progression of MDD. Strict selection and standardized processing ensured the national representativeness of the study sample, supporting the reliability and generalizability of the findings.

2.3. Assessment of depressive disorders

Depressive symptoms were assessed using the NHIS variable DEPLEVEL_A (Depression Severity), which includes values for: 1 = A little (mild), 2 = A lot (severe), 3 = Somewhere in between a little and a lot (moderate), 7 = Refused, 8 = Not Ascertained, 9 = Don't Know. The evaluation was based on the Patient Health Questionnaire-8 (PHQ-8), a widely validated tool for diagnosing depression and measuring its severity in clinical and population research (Wu et al., 2020). The PHQ-8 assesses symptoms over the past two weeks across eight items: low mood, lack of interest, sleep issues, fatigue, appetite changes, low self-esteem, concentration problems, and psychomotor agitation or retardation (Arias de la Torre et al., 2023). Each item is scored from 0 to 3 based on frequency, yielding a total score ranging from 0 to 24. Depression severity is classified as follows: 0–4 (No or minimal symptoms), 5–9 (Mild symptoms), 10–14 (Moderate symptoms), 15–19 (Moderately severe symptoms), 20–24 (Severe symptoms) (Toczyski and Feliksiak, 2024). A total score of ≥ 10 suggests clinically significant depressive symptoms (Fischer et al., 2023). If at least seven items were answered, missing values were imputed using the mean of completed items; otherwise, the status was classified as “unknown.” This comprehensive method accurately quantifies depressive symptom severity, providing a robust foundation for analyzing the key characteristics and progression of depressive disorders.

2.4. Statistical analysis

Data processing and analysis were performed using Python in the PyCharm 2023.1 environment (JetBrains, 2023). The study used NHIS data from 2019 to 2023, encompassing demographic characteristics, health status, and related variables. For feature selection, LASSO regression (L1 regularization) was applied to identify key variables and optimize model performance (El Guide et al., 2022). Prediction models were constructed using 13 machine learning algorithms, and model performance was evaluated based on precision, accuracy, recall, F1 score, and ROC-AUC. After standardizing all metrics, the best-performing model was selected. The best model underwent SHAP-based interpretability analysis, generating summary plots for the top 10 important features to clarify their influence on predictions. Through descriptive statistics, feature selection, model development, and interpretability analysis, this study systematically explores risk prediction for MDD and its key influencing factors, providing scientific evidence and data-driven insights for precision interventions and personalized management.

3. Results

3.1. Distribution of major depressive disorder

Between 2019 and 2023, the prevalence of moderate to severe

depression showed certain fluctuations. The incidence rates were as follows: 2019: 13.41 %, 2020: 11.98 %, 2022: 13.68 %, 2023: 13.19 %. Longitudinal analysis of the two cohorts revealed:

- 2019–2022 cohort ($n = 2688$): 309 individuals (11.50 %) experienced worsening depressive symptoms.
- 2020–2023 cohort ($n = 2754$): 319 individuals (11.58 %) experienced worsening depressive symptoms.

3.2. LASSO regression feature selection results

The number of publicly available variables in the NHIS database for 2019, 2020, 2022, and 2023 were 534, 617, 637, and 647, respectively. During the data processing stage, variable selection was conducted separately for the two main cohorts to ensure the completeness and consistency of study variables. For Group 1 and Group 3 (2019–2022 cohort), the shared variables numbered 424, and after removing columns with all missing values and the invalid variable HHX, 419 variables remained (removed variables: CHHDHP, GALLBAGETC, OGFLG, and CHFLG). For Group 2 and Group 4 (2020–2023 cohort), the shared variables numbered 444, and after similar cleaning, 439 variables remained (removed variables: CHHDHP, OGFLG, CHFLG, and PRPLCOV2). LASSO regression was further applied to refine the variables, determining the optimal λ values for the four models as 0.0005428675439323859, 0.004291934260128779, 0.010722672220103232, and 0.012328467394420659, respectively. The selected variables encompassed seven main domains: demographic characteristics, mental health, chronic and specific diseases, cognitive and physical functions, healthcare services and insurance, economic stress, and lifestyle factors, providing a multidimensional foundation for studying the onset and progression of depression. This variable selection process reflects the study's rigorous control over data quality and efficient extraction of significant features, facilitating a comprehensive analysis of key factors and mechanisms underlying depression. Details of the variables selected through LASSO regression for the four groups are presented in Table 1.

3.3. Prediction models and evaluation metrics for major depressive disorder risk

To ensure consistency across performance metrics (AUC, precision, recall, F1 score, etc.), scikit-learn's Standard Scaler was used to standardize all metrics, eliminating scale differences and allowing for a comprehensive total score (Total Score) to objectively compare and rank models. This framework enabled the selection of the most suitable model for each dataset and provided insights into model applicability and limitations in predicting depression risk. Results showed strong predictive performance in validation sets but significant declines in external validation, highlighting the impact of data distribution bias on generalizability. In Group 1 (2019 features predicting 2022 MDD risk), CatBoost achieved the best validation set performance (AUC: 0.8219, F1 score: 0.3797) but showed limited adaptability in external validation, whereas AdaBoost, with a Total Score of 0.8413 and a validation AUC of 0.9460, was chosen as the optimal model due to its robust performance across metrics. In Group 2 (2020 features predicting 2023 MDD risk), Gradient Boosting demonstrated outstanding performance with a validation AUC of 1.0000 and the highest Total Score (0.7889), outperforming AdaBoost and CatBoost despite their high AUCs of 0.9808 and 0.9771, respectively. For Group 3 (2019–2022 cohort predicting depression progression), Gaussian Naive Bayes (Total Score: 0.5214, AUC: 0.7595) excelled in identifying rare events with high recall (0.5876), making it suitable for capturing high-risk cases of progression. In Group 4 (2020–2023 cohort predicting depression progression), Extra Trees, with a validation AUC of 0.7579 and a Total Score of 0.5695, balanced precision and accuracy effectively, outperforming AdaBoost in overall scoring despite AdaBoost's slightly higher F1 score. This study

Table 1
Selected Features Across Four Cohorts for Depression Risk and Progression Prediction.

	Group 1	Group 2	Group 3	Group 4
Feature Selection	1. Demographic Characteristics Gender and Marital Status: SEX, MARITAL, MARSTAT, SPOUSESEX, SPOUSEP Age-related Variables: OVER65FLG, PCNT18UPTC, PCNTKIDS, HOUYRSLIV Race and Region: REGION, URBRL Education and Income: PRDEUC2, RATCAT, INCINTER, YRSINUS, POVRATTC Employment and Work: SPOUSWRK, HISTOPJOB, SOCWRKLIM 2. Health Status General Health: PHSTAT, WELLNESS BMI and Weight: BMICAT, WEIGHTLBTC Mental Health: Depression-Related: DEPEV, DEPFREQ, PHQCAT, PHQ84, PHQ81, PHQ82, PHQ83, PHQ85, PHQ86 Anxiety-Related: ANXEV, ANXLEVEL, ANXFREQ, GADCAT, GAD71, GAD72, GAD73, GAD74 Chronic Conditions: HYPEV, CHLEV, DIBINSSTYR, VAHOSP, ARTHEV Specific Conditions: CERVAGETC, LUNGAGETC, LIVERAGETC, SKNDKAGETC 3. Healthcare Utilization Visits and Preventive Care: DENPREV, WELLVIS, EYEEX12M Emergency and Hospital Visits: EMERG12MTC, URGNT12MTC, HINOTMYR Vaccinations: SHTFLUY, SHTPNEUNB, SHTFLUM, SHTTDPAP, SHTTETANUS Medication Use: CHLMED, DIBPILL 4. Lifestyle and Behavior Smoking: CIGNOW, SMKNOW, SMK30D, CIG30D, ECIGNOW Diet and Exercise: EQSTEPS, FDSLESS, FDSSSKIP, FDSHUNGRY Social Support: SOCWRKLIM, PARSTAT, PERASST 5. Economics and Insurance Economic Stress: PAYWORRY, PAYNOBLLNW Insurance: Private Insurance: PRVSCOV1, PRVSCOV2, PRHDHP1, PRHDHP2, HIKINDO2 Public Insurance: COVER65, POLHLD1, POLHLD2 6. Military and Special Experiences Military: MILSPC1, MILSPC2, COMBAT Life Events: SINCOVDE, HISTOPELIG, HISTOPJOB 7. Special Survey Sections Patient Health Questionnaire (PHQ): PHQCAT, PHQ84, PHQ81, PHQ82, PHQ83, PHQ85, PHQ86 Generalized Anxiety Disorder Scale (GAD): GADCAT, GAD71, GAD72, GAD73, GAD74 8. Special Variables Design and Stratification Variables: PSTRTAT, PCNTADTWKP, INT_QRT Housing and Family: HOUTENURE, HOUYRSLIV	1. Demographic Characteristics Marital Status: MARSTAT Residence and Housing: HOUYRSLIV, HOUTENURE Region: REGION Education: PRDEUC2, OPDEUC 2. Health Status Mental Health: Depression: DEPEV, DEPFREQ Anxiety: ANXEV, ANXLEVEL, ANXFREQ General Health: PHSTAT Chronic Conditions and Medication Use: Diabetes: DIBPILL Specific Conditions: CERVAGETC, PANCRAGETC, ESOPHAGETC Physical Functions: Cognitive Function: COGFRQDFF, COGMEMDFF, COGTYPEDFF Daily Activity Ability: NOEQWLK13M, EQWLK13M, EQSTEPS, NOEQSTEPS Pain and Health Limitations: Pain Frequency: PAIFRQ3M Health Limitations: REPLIMIT 3. Healthcare Services and Insurance Healthcare Utilization: Doctor Visits: INJSAWDOC, INJCHORES, ASAT12M Health Checkups: EYEEX12M, SHTFLU12M Vaccinations: TDAPPREG Insurance: Health Insurance Coverage: PRVSCOV1 Healthcare Cost Issues: RSNHICOST 4. Lifestyle and Behavior Diet and Physical Activity: Dietary Habits: FDSLESS, FDSSSKIP Physical Activity: EQSTEPS, NOEQSTEPS Smoking: Smoking History: SMKEV Work and Social Limitations: Social Work Limitations: SOCWRKLIM Work Limitation Causes: REPWRKCAUS 5. Economics and Family Economic Stress: Payment Issues: PAYWORRY, PAYNOBLLNW, PLNIPAY2 Family Status: PROXYREL, MCPARTD 6. Military and Special Experiences Military Experience: MILSPC2 Health Events: VIRAPP12M	1. Demographic Characteristics Region and Residence: REGION Education: MAXEDUCP, PRDEUC1, YRSINUS Family Structure: PCNTADLT 2. Health Status Mental Health: Depression-Related: PHQCAT, PHQ87, PHQ85, PHQ82, DEPEV, DEPMED Anxiety-Related: GADCAT, ANXFREQ Chronic Conditions and Physical Functions: Chronic Diseases: THYROAGETC, STOMAAGETC, LUNGAGETC, CVDSEV Cognitive Function: COGMEMDFF Pain: YOGAPAIN, MASSPAIN General Health: Activity Limitations: ASTILL Life Satisfaction: LSATIS4 3. Diet and Lifestyle Dietary Habits: Salad Consumption: SALADTP Food Shortage: FDSLESS, FDSSSKIP Long-Term Food Intake: FGELNGTRD Lifestyle: Activity Type: PLEXCHPR1 4. Healthcare Services and Vaccinations Vaccinations: SHTSHINGL1, SHTFLU12M Insurance Coverage: PRVSCOV2	1. Demographic Characteristics Work-Related: INJWRKDYTC, ARTHWRK Social Limitations: SOCWRKLIM Perceived Discrimination: DISCRIM1 2. Health Status Mental Health: Depression-Related: PHQ44, PHQ42, PHQ41, DEPEV Anxiety-Related: ANXMED, ANXFREQ Chronic Conditions and Pain: Chronic Diseases: SKNMAGETC, ARTHWRK, JNTPN Pain: PAIAPG3M Cognitive Function: Frequency: COGFRQDFF Type: COGTYPEDFF Memory: COGMEMDFF Physical Problems: Balance Issues: BFALLTIMES, BALDIZZ, CBALHDINJ Hearing Issues: AHEARST1 3. Diet and Lifestyle Diet: Food Shortage: FDSLESS Lunch Frequency: FLUNCH12M1 Physical Activity: Activity Vigilance: VIGIL4, VIGIL3 4. Healthcare Services and Insurance Healthcare Services: Emergency Services: URGNT12MTC Dental Services: DENPREV Vaccinations: SHTFLUY Insurance and Payment: Health Insurance: PRVSCOV1, MAHDHP Medical Bill Pressure: PAYNOBLLNW 5. Special Health Conditions Cancer Family History: FHBCAN50 Signs and Symptoms: SIGWHEN

Description: This table summarizes the variables selected through LASSO regression for the four study cohorts, spanning demographic characteristics, psychological health, chronic and specific diseases, cognitive and physical functions, healthcare utilization and insurance, economic stress, and lifestyle factors. These features form the foundation for analyzing major depressive disorder (MDD) risk and symptom progression, providing multidimensional insights into the critical factors driving MDD outcomes.

systematically compared machine learning models, revealing that Ada-Boost, Gradient Boosting, Gaussian Naive Bayes, and Extra Trees are suitable for different scenarios, providing a scientific framework for model evaluation and optimization for depression risk prediction. Detailed model metrics and standardized scores are presented in [Table 2](#), with validation set ROC curves illustrated in [Fig. 1](#).

3.4. Feature Importance analysis

This study employed SHAP (Shapley Additive Explanations) values to analyze model interpretability, revealing the contributions and mechanisms of important features in the best-performing models, and providing scientific evidence for identifying key factors in depression risk prediction. The top 10 most important features for each dataset

Table 2
Performance Metrics and Standardized Scores of Machine Learning Models.

	Model	AUC	Precision	Accuracy	Recall	F1	External AUC	External Precision	External Accuracy	External Recall	External F1	Standard AUC	Standard Precision	Standard Accuracy	Standard Recall	Standard F1	Total Score
Group 1	Logistic Regression	0.77	0.45	0.86	0.19	0.27	0.50	0.14	0.86	0.00	0.01	0.74	0.71	0.96	0.22	0.69	0.66
	Decision Tree	0.76	0.42	0.85	0.32	0.36	0.50	0.14	0.14	1.00	0.24	0.71	0.67	0.94	0.36	0.92	0.72
	Random Forest	0.80	0.63	0.88	0.13	0.21	0.60	0.19	0.78	0.19	0.19	0.88	1.00	0.99	0.14	0.53	0.71
	SVM (RBF Kernel)	0.74	0.57	0.87	0.07	0.12	0.50	0.00	0.86	0.00	0.00	0.59	0.91	0.98	0.07	0.30	0.57
	Gradient Boosting	0.81	0.56	0.88	0.29	0.39	0.57	0.14	0.14	1.00	0.24	0.96	0.90	0.99	0.33	0.99	0.84
	Extra Trees	0.80	0.59	0.88	0.14	0.22	0.60	0.00	0.86	0.00	0.00	0.89	0.93	0.99	0.15	0.57	0.71
	K-Neighbors (KNN)	0.67	0.37	0.86	0.12	0.18	0.50	0.00	0.86	0.00	0.00	0.24	0.59	0.95	0.13	0.45	0.47
	Gaussian NB	0.74	0.18	0.48	0.88	0.30	0.50	0.00	0.86	0.00	0.00	0.61	0.29	0.00	1.00	0.77	0.53
	AdaBoost	0.81	0.58	0.88	0.29	0.39	0.68	0.00	0.86	0.00	0.00	0.95	0.93	1.00	0.33	1.00	0.84
	MLP (Neural Network)	0.68	0.32	0.83	0.29	0.30	0.50	0.00	0.86	0.00	0.00	0.32	0.51	0.87	0.33	0.78	0.56
	XGBoost	0.79	0.51	0.87	0.28	0.36	0.71	0.00	0.86	0.00	0.00	0.86	0.81	0.98	0.32	0.92	0.78
	CatBoost	0.82	0.57	0.88	0.28	0.38	0.47	0.18	0.86	0.01	0.02	1.00	0.91	1.00	0.32	0.97	0.84
	Linear SVM	0.62	0.00	0.87	0.00	0.00	0.48	0.00	0.86	0.00	0.00	0.00	0.00	0.97	0.00	0.00	0.19
	Logistic Regression	0.76	0.56	0.88	0.06	0.10	0.74	0.00	0.87	0.00	0.00	0.83	0.71	0.98	0.12	0.30	0.59
	Decision Tree	0.71	0.41	0.86	0.29	0.34	0.50	0.00	0.87	0.00	0.00	0.70	0.53	0.84	0.61	1.00	0.74
Group 2	Random Forest	0.80	0.63	0.88	0.07	0.13	0.62	0.00	0.87	0.00	0.00	0.95	0.81	1.00	0.16	0.39	0.66
	SVM (RBF Kernel)	0.70	0.78	0.88	0.01	0.03	0.50	0.00	0.87	0.00	0.00	0.67	1.00	0.98	0.03	0.08	0.55
	Gradient Boosting	0.82	0.54	0.88	0.20	0.29	0.53	1.00	0.87	0.00	0.00	1.00	0.70	1.00	0.41	0.84	0.79
	Extra Trees	0.79	0.58	0.88	0.07	0.13	0.48	0.00	0.87	0.00	0.00	0.90	0.75	0.99	0.16	0.39	0.64
	K-Neighbors (KNN)	0.64	0.34	0.87	0.08	0.13	0.55	0.00	0.87	0.00	0.00	0.50	0.43	0.89	0.17	0.38	0.47
	Gaussian NB	0.74	0.25	0.77	0.48	0.33	0.50	0.20	0.87	0.00	0.00	0.78	0.33	0.00	1.00	0.96	0.61
	AdaBoost	0.81	0.54	0.88	0.20	0.29	0.63	0.00	0.87	0.00	0.00	0.98	0.69	1.00	0.41	0.84	0.79
	MLP (Neural Network)	0.66	0.28	0.83	0.27	0.27	0.57	0.22	0.85	0.06	0.09	0.57	0.36	0.52	0.56	0.79	0.56
	XGBoost	0.78	0.44	0.87	0.21	0.29	0.50	0.09	0.86	0.01	0.01	0.88	0.57	0.91	0.45	0.84	0.73
	CatBoost	0.81	0.54	0.88	0.17	0.26	0.58	0.00	0.87	0.00	0.00	0.98	0.69	0.99	0.36	0.76	0.76
	Linear SVM	0.46	0.00	0.88	0.00	0.00	0.44	0.00	0.87	0.00	0.00	0.00	0.00	0.97	0.00	0.00	0.19
	Logistic Regression	0.79	0.38	0.87	0.09	0.15						0.98	0.18	0.41	(0.30)	(0.08)	0.24
	Decision Tree	0.54	0.25	0.82	0.26	0.25						(2.72)	(0.76)	(1.45)	0.81	0.96	(0.63)
	Random Forest	0.78	0.44	0.88	0.04	0.08						0.89	0.71	0.62	(0.65)	(0.81)	0.15
	SVM (RBF Kernel)	0.72	0.50	0.88	0.02	0.04						(0.02)	1.13	0.66	(0.79)	(1.16)	(0.04)
Group 3	Gradient Boosting	0.75	0.42	0.87	0.11	0.18						0.52	0.55	0.50	(0.16)	0.22	0.32
	Extra Trees	0.77	0.57	0.88	0.04	0.08						0.70	1.67	0.71	(0.65)	(0.79)	0.33
	K-Neighbors (KNN)	0.65	0.40	0.87	0.12	0.19						(0.96)	0.37	0.41	(0.09)	0.32	0.01
	Gaussian NB	0.76	0.29	0.78	0.59	0.39						0.60	(0.46)	(2.85)	3.03	2.29	0.52
	AdaBoost	0.75	0.32	0.86	0.14	0.20						0.41	(0.24)	(0.01)	0.05	0.41	0.12
	MLP (Neural Network)	0.66	0.32	0.85	0.22	0.26						(0.86)	(0.24)	(0.35)	0.53	1.00	0.02
	XGBoost	0.73	0.31	0.87	0.09	0.14						0.13	(0.30)	0.20	(0.30)	(0.14)	(0.08)
	CatBoost	0.77	0.36	0.87	0.05	0.09						0.82	0.05	0.50	(0.58)	(0.66)	0.03

(continued on next page)

Table 2 (continued)

Model	AUC	Precision	Accuracy	Recall	F1	External AUC	External Precision	External Accuracy	External Recall	External F1	Standard AUC	Standard Precision	Standard Accuracy	Standard Recall	Standard F1	Total Score
Group 4	Linear SVM	0.69	0.00	0.88	0.00	0.00					(0.48)	(2.64)	0.66	(0.92)	(1.56)	(0.99)
	Logistic Regression	0.76	0.50	0.87	0.11	0.18					0.76	0.24	0.50	(0.09)	0.27	0.34
	Decision Tree	0.54	0.27	0.83	0.17	0.21					(1.84)	(0.70)	(0.95)	0.41	0.59	(0.50)
	Random Forest	0.75	0.55	0.87	0.05	0.10					0.70	0.43	0.55	(0.53)	(0.54)	0.12
	SVM (RBF Kernel)	0.67	1.00	0.87	0.01	0.02					(0.24)	2.30	0.55	(0.88)	(1.36)	0.07
	Gradient Boosting	0.75	0.55	0.87	0.11	0.18					0.65	0.43	0.59	(0.09)	0.29	0.37
	Extra Trees	0.76	0.73	0.88	0.10	0.18					0.76	1.20	0.82	(0.17)	0.24	0.57
	K-Neighbors (KNN)	0.65	0.19	0.85	0.04	0.06					(0.48)	(1.03)	(0.09)	(0.67)	(0.93)	(0.64)
	Gaussian NB	0.75	0.30	0.78	0.51	0.38					0.68	(0.59)	(2.93)	3.07	2.26	0.50
	AdaBoost	0.76	0.56	0.87	0.14	0.22					0.83	0.47	0.64	0.12	0.67	0.55
	MLP (Neural Network)	0.65	0.30	0.83	0.21	0.25					(0.50)	(0.57)	(0.86)	0.70	0.96	(0.05)
	XGBoost	0.72	0.41	0.86	0.08	0.14					0.29	(0.13)	0.32	(0.31)	(0.17)	0.00
	CatBoost	0.75	0.38	0.86	0.05	0.08					0.70	(0.23)	0.37	(0.60)	(0.72)	(0.10)
	Linear SVM	0.50	0.00	0.87	0.00	0.00					(2.30)	(1.82)	0.50	(0.96)	(1.55)	(1.22)

Description: This table presents the evaluation metrics (AUC, precision, recall, F1 score, and total score) and standardized scores of various machine learning models applied to the four study cohorts for predicting depression risk and symptom progression. The table highlights the best-performing models for each cohort (e.g., AdaBoost, Gradient Boosting, Gaussian NB, Extra Trees), providing a comparative analysis of model performance and their suitability for different predictive tasks. These results support the selection of optimal models for early screening and targeted interventions in major depressive disorder (MDD).

were visualized based on SHAP values, offering robust support for understanding model predictions and guiding intervention strategies. For Group 1 (2019 features predicting 2022 MDD risk), SHAP analysis identified key features such as PHQ82 (depression severity score), SHTFLU12M (flu vaccination in the past 12 months), and ANXLEVEL (anxiety level). Among these, PHQ82 contributed the most to the model's predictions, with SHAP values indicating that an increase in PHQ82 significantly elevated depression risk. The nonlinear impacts of SHTFLU12M and ANXLEVEL suggested complex interactions between biological health factors (e.g., vaccination) and psychological health indicators (e.g., anxiety). For Group 2 (2020 features predicting 2023 MDD risk), ANXLEVEL (anxiety level) continued to play a significant role in the model, followed by SHTFLU12M and DEPEV (experience of depressive episodes). SHAP values revealed that higher ANXLEVEL markedly increased depression risk, while the positive contribution of DEPEV aligned with the profound psychological burden associated with major depressive episodes. These findings underscore the joint influence of anxiety and depressive experiences in predicting depression risk. In Group 3 (2019–2022 cohort predicting depression progression), the most important features were COGMEMDF (cognitive memory deficit score), DEPMED (antidepressant use), and PCNTADLT (number of psychological counseling sessions). SHAP analysis showed that when COGMEMDF exceeded a certain threshold, the risk of depression progression increased significantly. The negative contribution of DEPMED suggested a protective effect of antidepressant treatment, while the positive effect of PCNTADLT highlighted the critical role of psychological interventions. For Group 4 (2020–2023 cohort predicting depression progression), SHAP analysis identified ANXFREQ (anxiety frequency), DEPEV, and DISCRIM1 (frequency of perceived discrimination) as the top features. High ANXFREQ values substantially increased the risk of worsening depression, while the positive impact of DISCRIM1 suggested that social stressors, such as experiences of discrimination, play a significant role in exacerbating depressive symptoms. These results, visualized in SHAP summary plots (Fig. 2), reveal the nuanced interplay between psychological, biological, and social factors in predicting depression risk and progression. The findings provide actionable insights for targeted interventions, emphasizing the importance of addressing both mental health symptoms and external stressors in depression management.

4. Discussion

4.1. Multidimensional predictive analysis of major depressive disorder incidence and progression

This study systematically analyzed the incidence, progression, and influencing factors of major depressive disorder (MDD) based on data from 2019 to 2023. The incidence rates for 2019, 2020, 2022, and 2023 were 13.41 %, 11.98 %, 13.68 %, and 13.19 %, respectively, demonstrating annual fluctuations. These variations may be attributed to changes in the socio-economic environment, the psychological health impacts of the COVID-19 pandemic, and shifts in the accessibility of mental health services(Kim et al., 2024). Longitudinal cohort analyses showed that the proportions of individuals with worsening depressive symptoms were 11.50 % in the 2019–2022 cohort and 11.58 % in the 2020–2023 cohort, indicating a relatively consistent risk of progression across different periods and samples. Using LASSO regression, the study identified key variables spanning seven major dimensions: demographic characteristics, mental health, chronic and specific diseases, cognitive and physical functions, healthcare and insurance, economic stress, and lifestyle factors. These multidimensional variables provide a robust foundation for exploring the factors influencing MDD onset and progression. The study further standardized model performance metrics and employed a comprehensive scoring system to compare multiple machine learning models, selecting the most suitable models for each dataset: AdaBoost for the first group, Gradient Boosting for the second

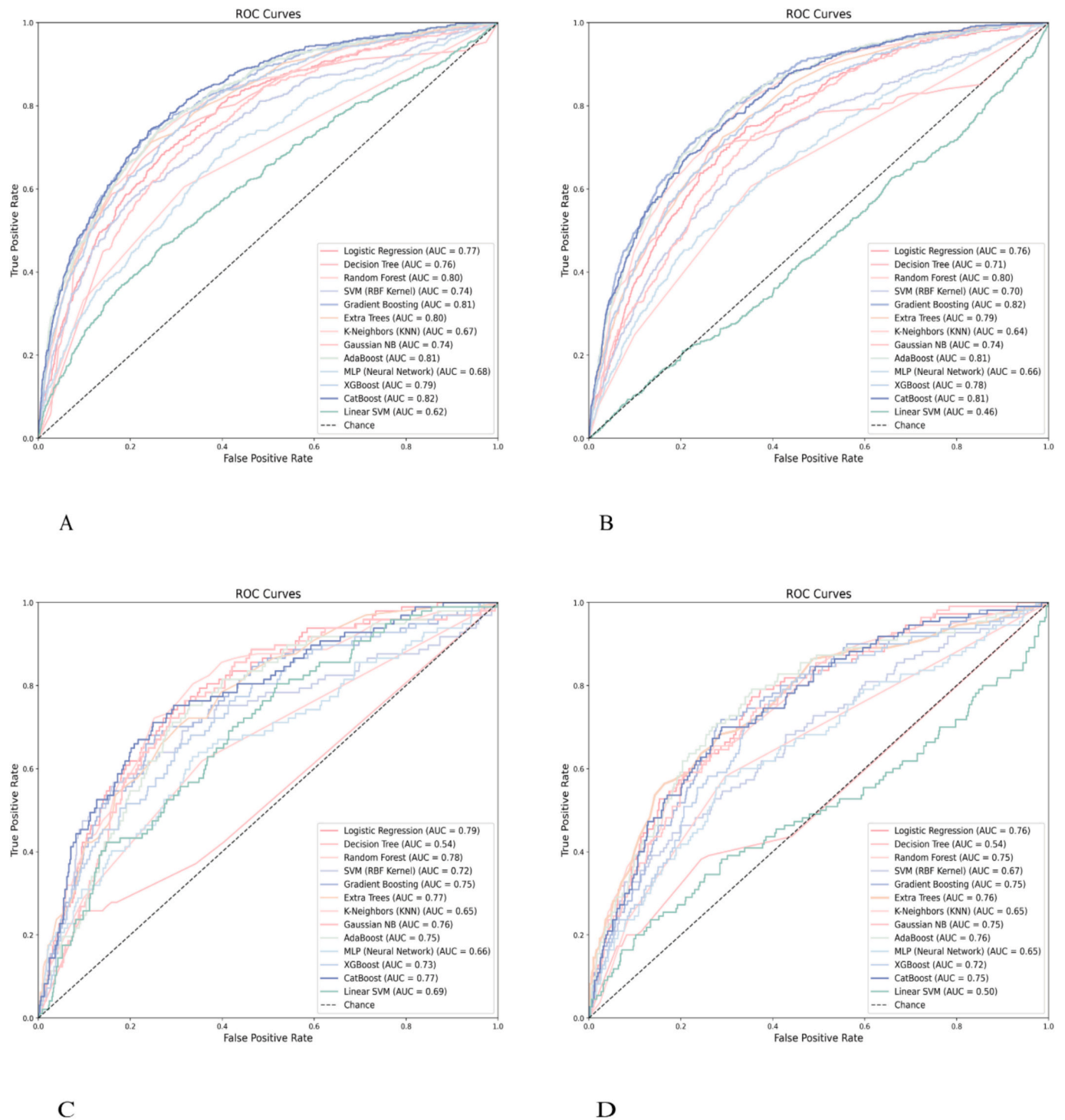


Fig. 1. ROC Curves of Machine Learning Models for Depression Risk and Progression Prediction.

Description:

This figure illustrates the ROC (Receiver Operating Characteristic) curves of machine learning models for the four study cohorts, highlighting their predictive performance in terms of true positive and false positive rates. Panel A represents Group 1 (2019 features predicting 2022 depression risk), Panel B represents Group 2 (2020 features predicting 2023 depression risk), Panel C represents Group 3 (2019–2022 cohort predicting depression progression), and Panel D represents Group 4 (2020–2023 cohort predicting depression progression). The curves demonstrate the models' discriminative capabilities and support the selection of optimal models for each predictive task.

group, Gaussian Naive Bayes for the third group, and Extra Trees for the fourth group. These findings offer valuable references for precise depression risk prediction across diverse scenarios. SHAP value analyses further elucidated the contributions and mechanisms of features in the best-performing models. In the first dataset, PHQ82 (frequency of

feeling down, depressed, or hopeless in the past two weeks) emerged as the most important feature, highlighting the direct association between depressive scores and depression risk. In the second dataset, ANXLEVEL (anxiety level) and DEPEV (experience of depressive episodes) stood out as critical contributors, emphasizing the comprehensive impact of

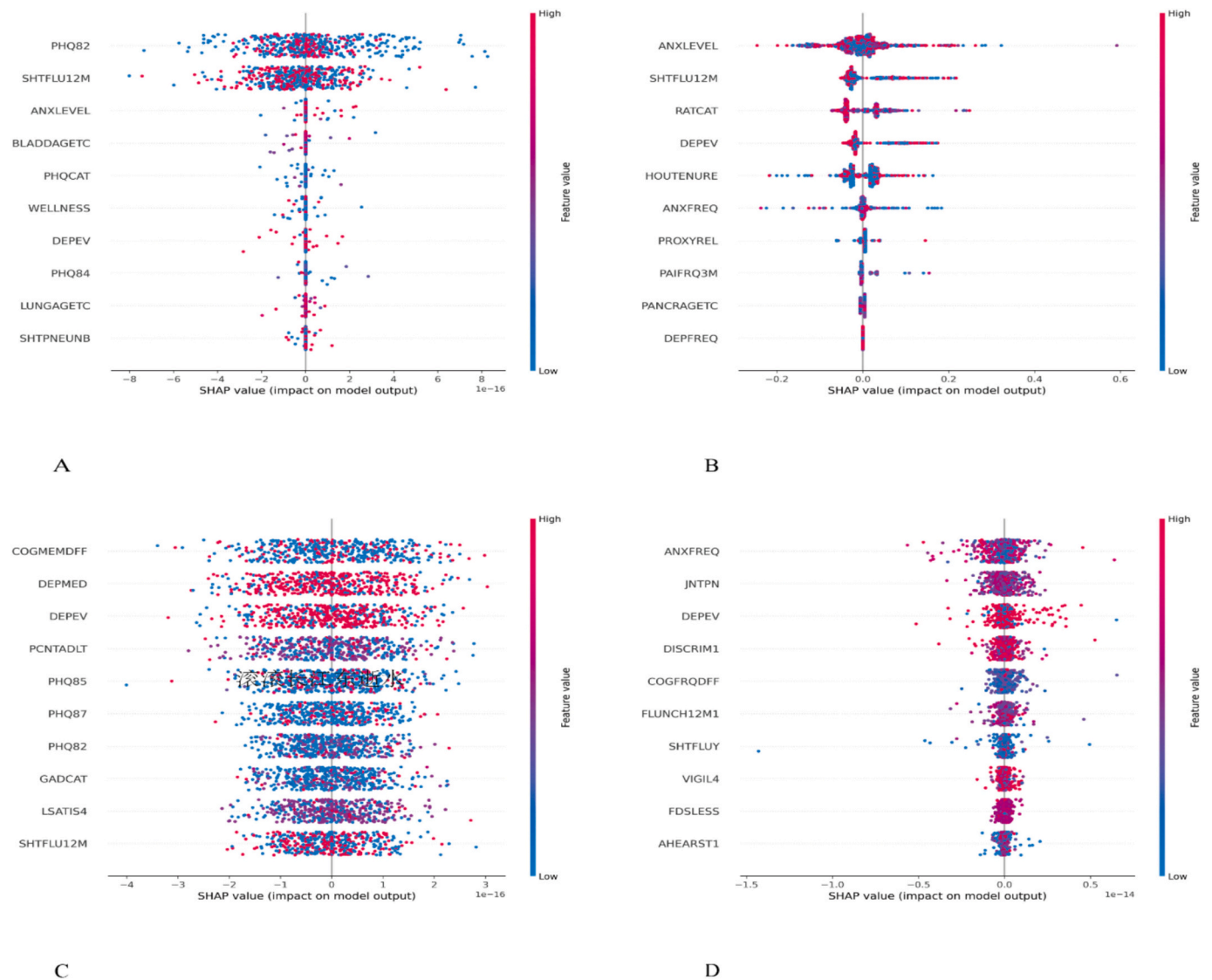


Fig. 2. SHAP Summary Plots of Key Features in Predictive Models for Depression Risk and Progression

Description:
This figure presents SHAP (Shapley Additive Explanations) summary plots for the top 10 important features identified in the best-performing models across the four study cohorts. Panel A corresponds to Group 1 (2019 features predicting 2022 depression risk), Panel B to Group 2 (2020 features predicting 2023 depression risk), Panel C to Group 3 (2019–2022 cohort predicting depression progression), and Panel D to Group 4 (2020–2023 cohort predicting depression progression). The plots quantify the contribution of each feature to model predictions and illustrate the influence patterns of key variables such as PHQ82, ANXLEVEL, and DISCRIM1, providing critical insights into the mechanisms underlying major depressive disorder (MDD) risk and progression.

mental health. In the third and fourth datasets, COGMEMDF (cognitive deficit score), PCNTADLT (number of psychological counseling sessions), ANXFREQ (frequency of anxiety), and DISCRIM1 (frequency of perceived discrimination) were identified as core predictors of symptom progression and worsening. These results underscore the importance of mental health, social support, and biological health in predicting depression. They provide crucial data-driven insights to inform intervention and management strategies for MDD, offering a foundation for targeted and effective support for at-risk populations.

4.2. Comparative analysis with existing research and innovative contributions

Compared to existing studies, this research demonstrates significant advantages in data scope, feature selection, and methodological applications, while further expanding the potential of depression risk prediction in the general population. Unlike the work of Yiye Zhang, which

focused on postpartum depression prediction using electronic health records, this study utilized a broad population dataset(Zhang et al., 2024). By combining LASSO regression for variable selection with SHAP value analysis, it identified core factors such as anxiety level (ANXLEVEL), depression score (PHQ82), and social support variables (DISCRIM1, PCNTADLT) as critical contributors to depression onset and progression. These findings(Ai et al., 2024; Wang et al., 2024) align with Yu Wang’s emphasis on long-term monitoring of postpartum depression and Fangzhu Ai’s analysis of depression risk among the elderly. However, this study extends the analysis to a wider age range and multiple timeframes, offering a more comprehensive understanding of depression risks. Methodologically, this study adopted a systematic evaluation framework for machine learning model comparison, selecting the best-performing models (e.g., AdaBoost, Gradient Boosting, and Gaussian Naive Bayes) for different tasks, providing scientific support for model application. This approach addresses the limitations of Radwan Qasrawi, whose research on childhood depression relied on Random Forest

and Support Vector Machine, by introducing diverse models and employing SHAP values for interpretability (Qasrawi et al., 2022). The findings revealed the critical role of social support, cognitive function, and health behavior variables in depression prediction. In comparison with Nikolaos Koutsouleris, who utilized multimodal data such as neuroimaging and genetic risk scores for prediction (Koutsouleris et al., 2018), this study focused on readily accessible social, psychological, and behavioral variables. This choice enhances the model's generalizability and practical applicability. However, the lack of integration with biological and imaging data may limit the depth of insights into the mechanisms underlying depression. Additionally, the decline in model performance during external validation highlights challenges related to data distribution shifts, necessitating further optimization. This study also prioritized depression risk assessment across multiple age groups rather than focusing on specific populations, such as postpartum women or the elderly. While this broad approach provides a holistic view, it leaves room for improvement in precision targeting for specific populations. By systematically comparing models and analyzing features, the study revealed multidimensional factors influencing depression risk and explored predictive scenarios in the general population. Compared to existing research, this study expands the application scenarios of depression prediction models while providing a scientific framework for model selection and interpretability. It offers critical insights for optimizing model performance and clinical applications in practice. Future research could incorporate multimodal data integration, targeted analyses for specific populations, and external validation strategies to advance depression risk prediction toward greater precision and diversity.

4.3. SHAP variable analysis and multidimensional exploration of MDD mechanisms

The variables extracted through SHAP analysis reveal the multilayered mechanisms underlying major depressive disorder (MDD) risk, encompassing psychological health, biological characteristics, and social environmental dimensions. These variables are not only crucial for depression risk prediction but also reflect the complex dynamics of MDD onset and progression. Among psychological health variables, PHQ82 (an item assessing the frequency of feeling down, depressed, or hopeless over the past two weeks) stands out as a core indicator of depression severity (Schlechter et al., 2024). It highlights the direct relationship between depressive symptoms and functional impairments, particularly prefrontal cortex dysfunction (Pizzagalli and Roberts, 2022). High PHQ82 values significantly elevate depression risk, reinforcing its utility in identifying individuals with severe depressive symptoms. Similarly, ANXLEVEL (anxiety level) underscores the neurobiological comorbidity between anxiety and depression, involving mechanisms such as hyperactivation of the HPA axis and dysregulated neurotransmitters (e.g., serotonin and norepinephrine) (Kupcova et al., 2022). These mechanisms independently influence depression risk and may exacerbate stress responses. In terms of social support and environmental variables, DISCRIM1 (perceived discrimination) and PCNTADLT (number of psychological counseling sessions) illustrate opposing mechanisms. Discrimination can activate chronic stress responses and weaken social support networks, intensifying feelings of isolation and helplessness, which aggravate depressive symptoms (Lee et al., 2022). Conversely, psychological counseling provides emotional support, improves cognitive strategies, and enhances coping abilities, mitigating symptom progression (Kopelovich et al., 2021). Biological characteristics and health behavior variables, such as COGMEMDF (cognitive memory deficit score), highlight the bidirectional relationship between depression and cognitive dysfunction. Depressed individuals often exhibit reduced executive function and memory deficits, which may stem from abnormalities in the prefrontal cortex and hippocampus, further compounding depressive symptoms (Nusslock et al., 2024). Interestingly, health behavior variables like SHTFLU12M (flu vaccination) are not directly

tied to the biological mechanisms of depression but serve as proxies for health management behaviors. These behaviors may reflect social interaction frequency and health awareness, indirectly influencing mental health outcomes (Howard-Jones et al., 2022). SHAP values also reveal potential interactions among variables. For instance, ANXFREQ (anxiety frequency) and PHQ82 may mutually reinforce each other, collectively elevating depression risk. Interactions between social support (e.g., psychological counseling) and health behaviors (e.g., vaccination) could further influence individuals' resilience to life stressors. Understanding these mechanisms helps illuminate the complex etiology of MDD and provides actionable directions for early intervention and precision management. For example, identifying individuals with high anxiety or discrimination experiences can aid in early detection of high-risk groups. Strengthening cognitive interventions and building psychological support networks may effectively slow depression progression. Future studies could integrate neuroimaging, biomarkers, and social behavioral data to explore complex variable interactions and provide a more comprehensive foundation for MDD prediction and management.

4.4. Study limitations

Despite uncovering key factors influencing MDD through multidimensional feature analysis and machine learning model evaluation, this study has several limitations. First, the data primarily come from the U. S. National Health Interview Survey (NHIS), which may limit the generalizability of models due to the geographic and cultural specificity of the sample. The limited time span also hinders capturing the long-term trends and impacts of depression. Second, core variables like PHQ82 and ANXLEVEL rely on self-reported questionnaires, which are subject to subjective bias. Simplified measurements of social support and environmental variables (e.g., DISCRIM1, PCNTADLT) may underestimate their contributions. Although models performed well on validation sets, performance declines in external validation suggest challenges related to data distribution differences and overfitting. Complex models (e.g., Gradient Boosting and Extra Trees) also have limitations in interpreting feature interactions. Additionally, predicting rare events such as depression progression and worsening remains challenging due to data imbalance. Lastly, this study employs predictive analysis methods, which do not directly uncover causal relationships among variables. Future research could combine causal inference models to further explore underlying mechanisms. These limitations suggest directions for improvement, including expanding data scope, optimizing models, refining feature measurement tools, and incorporating causal inference methods. Such advancements could enhance the precision and applicability of depression risk prediction.

5. Conclusion

This study systematically analyzed data from 2019 to 2023, revealing trends in the incidence of major depressive disorder (MDD), characteristics of symptom progression, and multidimensional influencing factors. The findings indicate that depression risk is closely associated with key variables related to psychological health (e.g., PHQ82, ANXLEVEL), social support (e.g., DISCRIM1, PCNTADLT), and biological health (e.g., COGMEMDF, DEPEV). Using LASSO regression for feature selection and evaluating multiple machine learning models, the study identified optimal models for each dataset (AdaBoost, Gradient Boosting, Gaussian Naive Bayes, and Extra Trees), demonstrating strong predictive performance across different tasks. SHAP value analysis further quantified model interpretability, elucidating the influence patterns of critical features on depression risk. The results highlight that psychological health variables (e.g., depression score and anxiety level) are central to depression risk prediction, while social support and cognitive deficits play crucial roles in the progression and worsening of symptoms. This research provides scientific evidence for

early screening of depression risk, the development of personalized intervention strategies, and the optimization of public health policies. However, limitations in data scope and model applicability remain. Future efforts should focus on expanding data coverage, optimizing models, and incorporating causal inference methods to enhance the precision and generalizability of depression risk prediction. This study offers important theoretical support and practical guidance for the prediction, management, and intervention of depression.

CRedit authorship contribution statement

Yubei Lin: Conceptualization, Data curation, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Chuang Li:** Visualization, Writing – original draft. **Hongyu Li:** Data curation, Supervision, Writing – original draft.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT4o in order to translation. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare no conflict of interest related to this study. [If applicable, provide specific details about financial or non-financial interests that might influence the research or its outcomes.]

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Data availability

Data will be made available on request.

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